

## Filtering – Simple Explanation

```
import numpy as np

import matplotlib.pyplot as plt

from scipy.signal import butter, filtfilt

# Sampling frequency

fs = 1000 # Hz

# Time signal

t = np.arange(0, 1, 1/fs)

# Original signal: 50 Hz + 300 Hz noise

signal = np.sin(2 * np.pi * 50 * t) + 0.5 * np.sin(2 * np.pi * 300 * t)

# Low-pass filter

cutoff = 100 # Cutoff frequency in Hz

order = 4

# Design Butterworth filter

b, a = butter(order, cutoff / (fs / 2), btype='low')

# Apply filter

filtered_signal = filtfilt(b, a, signal)

# Plot original signal

plt.figure(figsize=(10, 6))
```

```
plt.subplot(2, 1, 1)
plt.plot(t, signal)
plt.title("Original Signal")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid()
```

```
# Plot filtered signal
plt.subplot(2, 1, 2)
plt.plot(t, filtered_signal)
plt.title("Filtered Signal - Low Pass Filter")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid()
```

```
plt.tight_layout()
plt.show()
```

OUTPUT:

